Uber Price Elasticity via NYC Congestion Pricing

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2025 Congestion Pricing Policy

01 - Policy

CRZ Toll Goes Live

As of Jan 5, 2025 NYC implemented congestion pricing.

All vehicles entering Manhattan below 60th street (excluding FDR & West Side hwy) are tolled.

02 - Rates

Pricing Structure

$9 - Private Cars

$1.50 - Rideshare (e.g. Uber)

$0.75 - Taxis

03 - Response

Pass Fee to Riders

Uber passes $1.50 surcharge directly to passengers, and was an early lobbyist for the policy.

04 - Question

What Can We Learn?

Using data from before and after NYC implemented this toll, we aim to determine Uber's price elasticity and whether they are under-charging customers.

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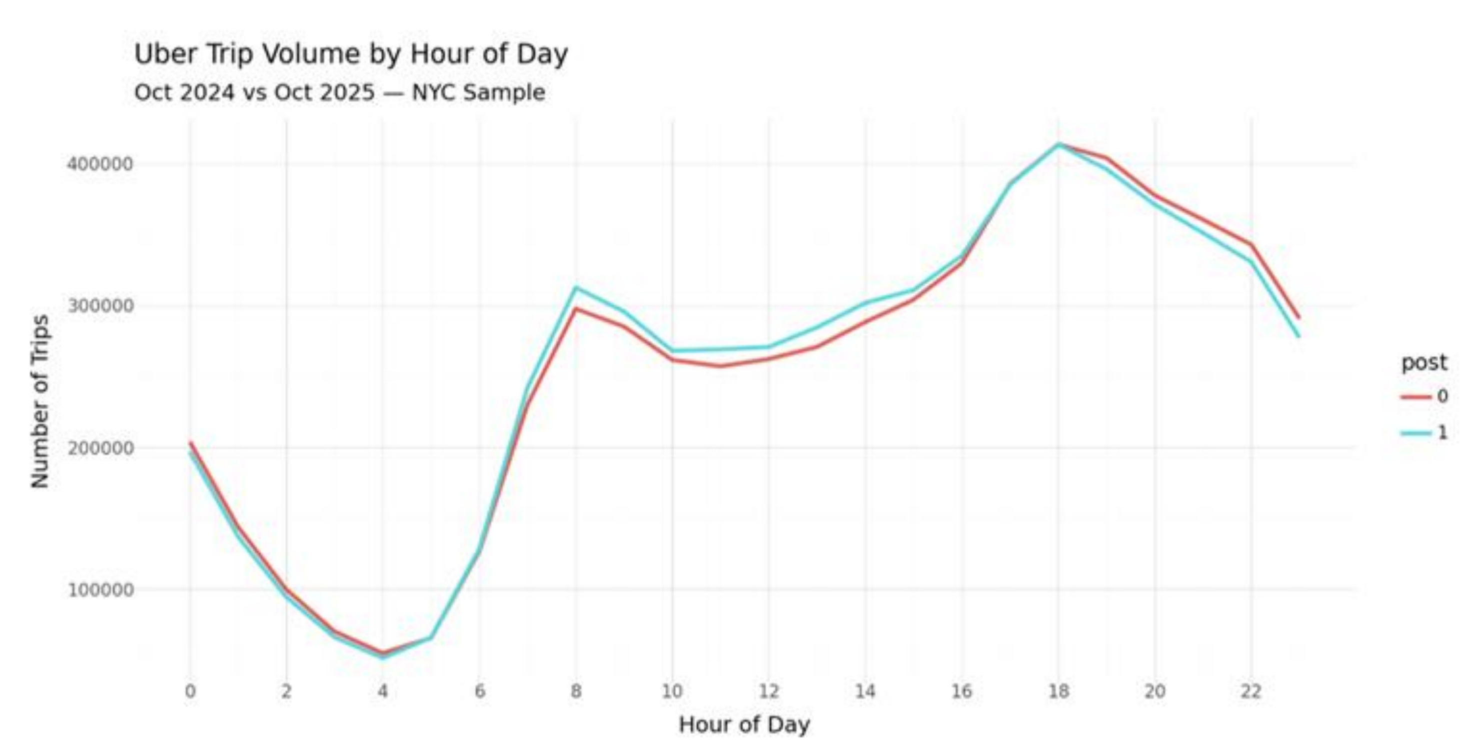
Policy Outcomes on Demand & Price



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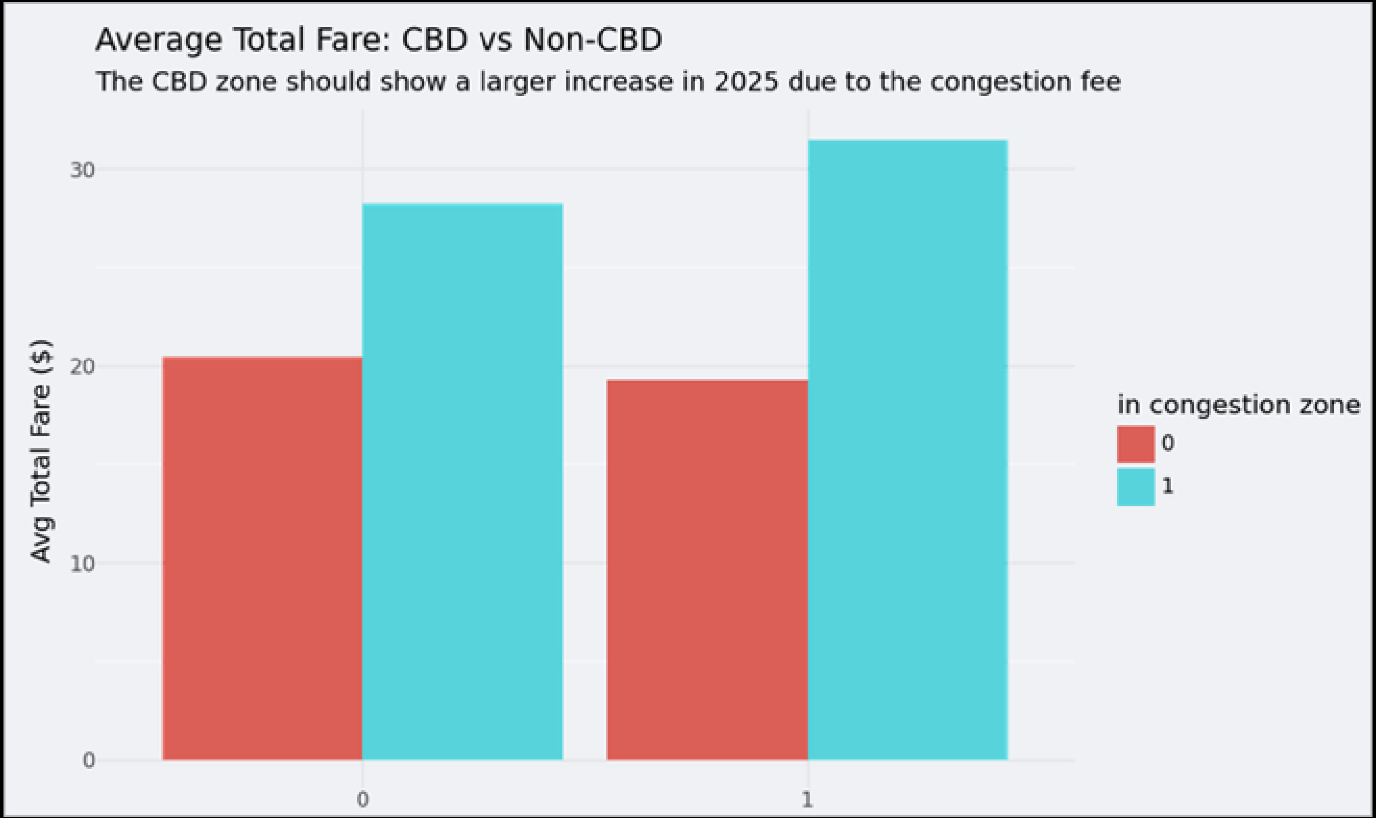
Policy Outcomes on Demand & Price



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Can We Predict Demand for Every % Change in Trip Price?

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Deriving Elasticity

01 - Study Design

Natural Experiment

Congestion pricing policy used as exogenous instrument.

Policy shock creates clean variation in price not driven by demand.

Avoids endogeneity bias present in standard OLS estimates.

02 - Data Window

Pre & Post Comparison

Oct 2024 vs Oct 2025 as pre and post policy observation periods.

October selected to minimize external noise: no major holidays, low tourism seasonality, no large events, and enough time elapsed since law passed.

Isolated month reduces confounding relative to other periods.

03 - Data Cleaning

Cleaned Dataset

Raw trip-level data processed and filtered for quality.

Consistent zone definitions applied to congestion area with filtering to isolate Manhattan and Brooklyn.

Removed cross-zone trips (i.e. you start inside the CRZ and end up outside), data discrepancies, transport subsidies, and negative base fair values.

04 - Estimation

2SLS Regression

Two-Stage Least Squares uses policy as instrument for price.

Stage 1: predict price change from congestion policy assignment.

Stage 2: estimate causal price elasticity of rideshare demand.

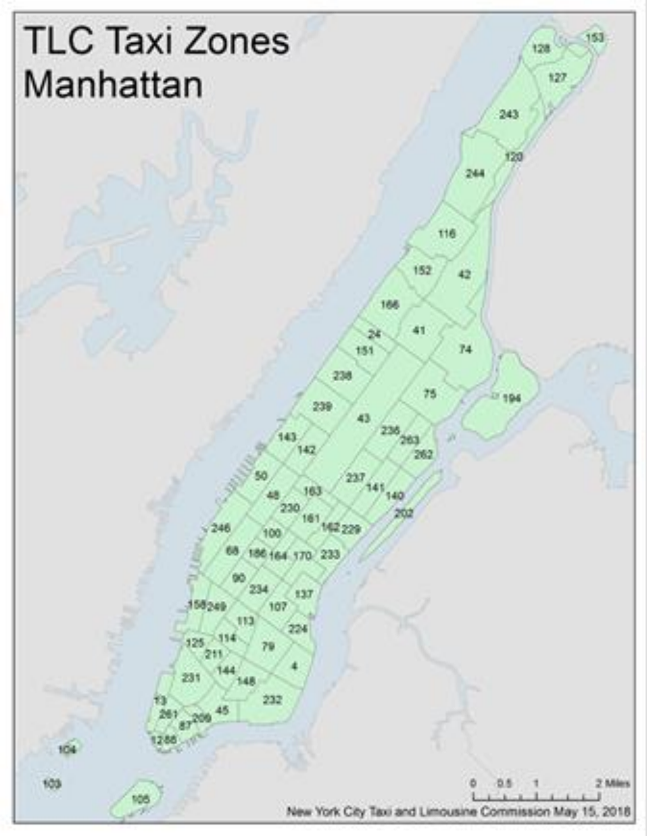
Data Preparation

Data Sources:

- **TLC Trip Record Data** (30+ columns, 1M+ records)
- **Zone ID metadata** (Region, Borough, etc.., ~30 records)

Steps:

- **Filter records in Oct 2024 & 2025 only** (flag pre/post)
- **Enrich trip records with Zone metadata** (Join)
- **Flag in-congestion zone trips & remove mixed-zone trips**
- **Remove negative prices & inapplicable trips**
- **Identify discrepancies & form clean data buckets**



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Data Cleaning

Discrepancies:

- **Congestion Fee applied or skipped contrary to expectation**

(Percent of trips with fee applied)

	Post Congestion
Inside Zone	99.83%
Outside Zone	2.73%

Remove outliers

post	in_congestion_zone	mean_cbd_fee	n_obs	mean_price
i32	i32	f64	u32	f64
0	0	0.0	3847337	20.458857
0	1	0.0	2276100	28.243145
1	0	0.0	3903255	19.312346
1	1	1.5	2249080	31.45851

Regression

Aggregation

- Aggregate trips into buckets (PULocation x hour x date)
- log-transform price & demand

PULocationID	date	hour	post	in_congestion_zone	avg_price	avg_gross_margin	log_Q	log_P	DID
str	date	str	str	i32	f64	f64	f64	f64	i32
"54"	2025-10-03	"23"	"1"	0	22.736154	2.634615	2.564949	3.123956	0
"225"	2024-10-13	"18"	"0"	0	17.255235	2.210765	5.135798	2.848116	0
"11"	2025-10-07	"12"	"1"	0	17.386429	2.576429	2.639057	2.85569	0

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Regression

Aggregation

- Aggregate trips into buckets (PULocation x hour x date)
- log-transform avg price & demand

log(P) ~ date + hour + location + (post x in-zone)

```
Linear regression (OLS)
Data                   : panel
Response variable      : log_P
Explanatory variables: date, hour, DID, PULocationID

R-squared: 0.583, Adjusted R-squared: 0.582
F-statistic: 1197.712 df(215, 184223), p.value < 0.001
Nr obs: 184,439
```

log(Q) ~ date + hour + PULocationID + log(P)

```
Linear regression (OLS)
Data                   : panel
Response variable      : log_Q
Explanatory variables: date, hour, PULocationID, predicted_log_price

R-squared: 0.831, Adjusted R-squared: 0.831
F-statistic: 4223.378 df(215, 184223), p.value < 0.001
Nr obs: 184,439
```

Predicted prices

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Main Result

Elasticity (ε) = –0.135

i.e. 1% change in price results in 0.135% decrease in trips

This indicates highly inelastic demand; short-run pricing power in Manhattan.

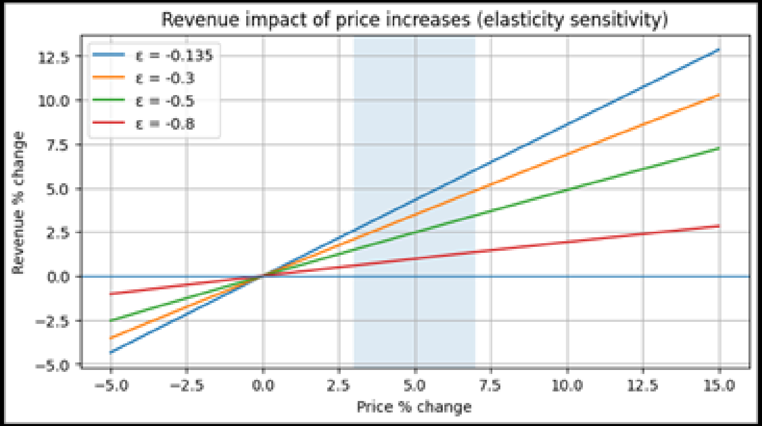
Uber Manhattan	*-0.135*
Gasoline (short-run)	*approx. -0.2*
Airline Tickets	*approx. -1.2*

Our findings suggest that a price increase may be viable.

Deriving Optimal Price

Theoretical optimal markup: $-1/\varepsilon$ = 741%

This is clearly unrealistic, and may indicate that elasticity is short-run only or Uber is heavily constrained by competition or regulation.



- Revenue increases for moderate price increases.
- Testing under more conservative elasticity (-0.3, -0.5, and -0.8) for robustness, we find that revenue rises in at least 3-7% range.
- Indicates short-run revenue pricing power.

Assuming Uber wants to increase profit but avoid >2% drop in rides, we find:

- Optimal 16.14% price increase
- Yields 13.82% revenue increase

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Pricing Recommendation

Main Recommendation: Implement a targeted 3-7% base fare increase in Manhattan congestion zone.

Rationale: demand is inelastic, revenue and profit will increase within tested range of elasticity values; customer willingness to pay not captured by current pricing.

Execution:

- Congestion-related pricing update
- Roll out via controlled A/B testing
- Monitor trip volume and competitive response

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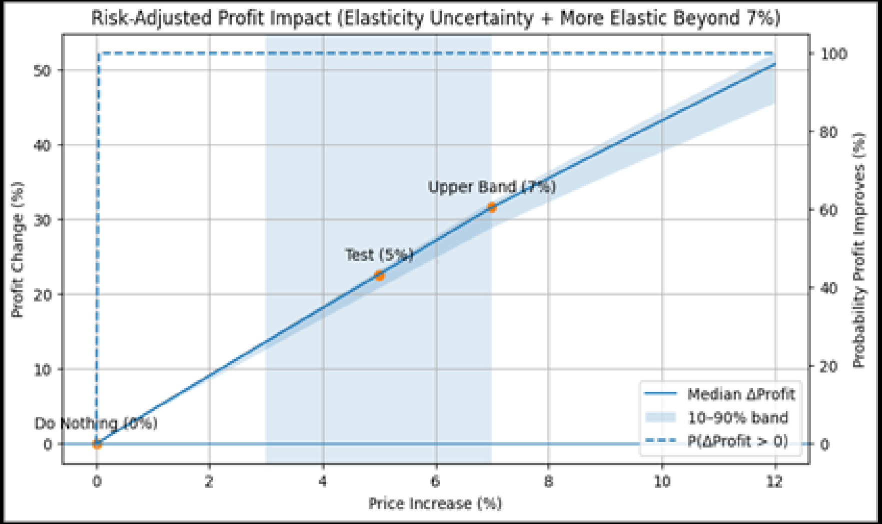
Alternative Strategies

Two Alternative Approaches:

- Maintain baseline (0% increase)
- Larger price increase (>7%)



Baseline: Risk adjusted assessment of profit impact indicates that maintaining the baseline with 0% change leaves profit on the table given the inelastic demand.

Larger Increase: May trigger stronger substitution effects outside of the local elasticity measured from the congestion surcharge.

This strategy would incur higher levels of risk and expose Uber to political/public backlash and opportunities for competitors to undercut.

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Key Takeaways

Short-run Elasticity in Manhattan: **(ε) = –0.135** (highly inelastic)

Theoretical Optimal Price Increase:

Constrained to <2% decrease in number of rides

- 16.14% price increase
- 13.82% revenue increase

Core Recommendation: implement a targeted 3-7% base fare increase in congestion zones as a congestion pricing update using controlled experimentation.

Limitations: data reflects short-run local variation; larger price changes may induce unknown or stronger effects.

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Future Considerations

01 - Methodology Extension

Heterogeneous elasticity by trip type:

Short trips (<2mi) in Manhattan are likely more elastic (easy to walk, subway, bike) than long trips. Splitting by trip length could reveal where Uber has the most vs. least pricing power.

02 - Business Question

Lyft comparison:

Running our same methodologies on Lyft data could show whether demand shifted between platforms rather than just disappearing.

03 - Policy Implications

Fare exemptions and their impact:

The MTA exempts certain trip types from the full $9 fee (shared rides, WAV vehicles). Analyzing whether riders shifted toward exempt trip types post-policy would reveal substitution within Uber's own product lineup.